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Is the Best, the Best? Analysing Best-Paper Awardees' Career Trajectories

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1 Introduction

At pretty much every computer science conference, a handful of papers get singled out as “best.” Committees deliberate, some teams walk up on stage, and everyone adds a new line to their CV. But what actually happens after that? Does winning change anything for the junior researchers involved, or does it mostly just confirm what was already going to happen anyway.

Recognition matters a lot in how science works [1, 2]. Tenure decisions, grant competitions, hiring all of these lean heavily on signals like publications, citations, and prizes [3, 4]. And informally too, an award changes how people read your work, how likely collaborators are to reach out, how your application looks in a pile [5, 6, 7]. But the relationship between getting recognised and what comes next is not obvious. Studies of big prizes like the Nobel or Fields Medal show that winning can sometimes be followed by lower citation impact per paper, as though being celebrated changes what people choose to work on [8, 9]. Other work finds the opposite, that prize winners publish more and stay active longer than matched peers who did not win [10, 2, 11]. So awards might be both a marker of past excellence and something that actually reshapes what comes next.

This raises questions about fairness that matter beyond any individual career [5, 1]. If early recognition accelerates some careers while others with similar potential never get that push, then awards might be amplifying inequalities more than just recording achievement. Or maybe the effect is small and awards mostly just find people who were already on a good path. Either way it matters not just for the researchers themselves but for universities, funders, and conference organisers who use these awards in evaluation all the time.

Most of the research on this looks at rare, high-profile prizes for already established scientists. Conferences are where most of the field publishes, and best paper awards happen far more often than global prizes. They often go to teams with PhD students or recent graduates, right when careers are still being shaped. For a junior researcher, a best paper award might be the first real signal that their work is landing. But we do not actually know much about whether those awards are followed by meaningfully different career paths. Do junior awardees go on to publish more, get cited more, do more disruptive work or do their careers look about the same as similar people who just did not happen to win.

This thesis tries to answer whether winning a best-paper award at a CS conference measurably changes the career trajectory of the junior researchers involved, compared with equally promising peers who did not win. It looks at the career trajectories of junior authors who won best paper awards at CS conferences and compares them with matched non-winning peers from the same venues and years. “Junior” here means within the first five years of a researcher’s indexed publication record, the window in which early-career

factors such as visibility, collaboration access, and initial citation accumulation have been shown to largely determine long-run research impact [12, 13, 10]. By “career trajectory” I mean the bibliometric indicators that are measurable at scale, annual publication counts, citation accumulation, and a disruption score based on the CD index (Section 2.4), which captures whether a paper redirects or consolidates the literature [14, 15]. These are imperfect proxies for quality or creativity, but they are the signals on which hiring committees, grant panels, and tenure decisions demonstrably rely [3, 4, 5].

Work on early-career recognition suggests that getting prizes earlier tends to amplify future achievement, more papers, more citations, longer active careers, even after accounting for initial performance [10]. Research on what predicts long-term impact shows that a small set of early-career features (like citations, institutional prestige, senior collaborators) explain a lot of variation across fields [12]. And work on team structure shows that small teams are more likely to produce disruptive work while larger teams tend to consolidate existing lines [14]. Any effect of conference awards on junior researchers is probably tangled up with all of these things.

Against that background, the thesis asks three main questions. First, do the publication and citation trajectories of junior awardees differ from those of similar non-awardees from the same conferences and years, and if so, in which direction. Second, do awardees and non-awardees diverge in how disruptive their subsequent work is, using citation-based indices that capture whether papers redirect or consolidate the literature. Third, are any differences consistent across venues, or concentrated in particular contexts specifically, whether effects on junior awardees vary across venue prestige tiers as measured by the CORE ranking system.

The data come from OpenAlex¹ combined with curated best paper award lists. By comparing awardees with carefully matched non-awardees, the thesis aims to give an empirical picture of what conference-level recognition actually does to junior researchers’ careers in CS and what that might mean for how we design and interpret these recognition systems.

2 Related Work

2.1 Scientific Recognition and the Reward System in Science

A well-known idea in the sociology of science is the *Matthew effect*, named after the Gospel of Matthew (‘to him who has, more will be given’), the observation that scientists who are already well-known, tend to get more credit for their work than lesser-known researchers doing equally good work [1]. In practice, this means that small early advantages in a research career can compound over time into large differences in visibility, funding, and

¹OpenAlex is a bibliographic catalogue of scientific papers, authors, and institutions.

prestige.

One of the clearest empirical demonstrations of this comes from [5], who looked at a Dutch research funding programme and found that scientists who just barely won a grant went on to attract more than twice as much follow-up funding compared to scientists who just barely missed out, despite being nearly identical in quality at the time of the decision. Similar compounding dynamics have been documented at the level of individual careers, where early citation and productivity advantages consistently predict long-term research impact [1, 2]. This kind of threshold effect suggests that formal recognition can have long-lasting consequences that go well beyond the award itself, which is exactly what motivates looking at best paper awards in this thesis.

2.2 Best Paper Awards in Computer Science

Computer science is unusual among academic disciplines in that conferences, not journals, are the main venue for publishing research, partly because the field moves fast enough that the year-long turnaround of journal review cycles is poorly suited to its publication norms [16]. Best paper awards at these conferences are therefore a meaningful signal of quality and visibility within the community [17]. Huang et al. [18] compiled a dataset of best paper awards across 32 major CS conferences going back to 1996, which is the starting point for this thesis. Beyond citation impact, awards also function as non-monetary incentives that confer social status and public recognition, shaping how recipients are perceived within their community [19].

Some prior work has looked at the citation impact of award-winning CS papers. [17] found that papers selected for awards by programme committees before publication tend to accumulate more citations than those selected after publication. But most of this work focuses on the papers themselves. What is much less studied is what happens to the *people* who wrote those papers, especially the junior authors who may have done most of the work but are just getting started in their careers. That gap is one of the things this thesis tries to address.

2.3 Junior Authors and Early Career Trajectories

Getting the right visibility early in an academic career seems to matter a lot. [6] showed that the prestige of a researcher’s doctoral advisor acts as a ‘chaperone’, directly shaping their placement and long-term career outcomes independent of their own output. Along similar lines, [7] found that junior researchers who co-authored with top scientists early in their careers consistently outperformed peers with similar ability levels who did not have that exposure. Interestingly, this effect was biggest for researchers who were not already at elite institutions, suggesting that external signals of quality can substitute for prestige

in some cases.

While conferences distribute various forms of recognition, best student paper, honourable mentions, test-of-time awards, best paper awards are the most consistently awarded across venues and the most widely documented in curated datasets [18], making them the most tractable for large-scale empirical analysis.

Does being on a best paper at a major CS conference do something similar for junior authors? It puts their name in front of the whole community, attached to work that has been officially recognised as excellent. Whether that translates into more publications, more citations, or better career opportunities remains an open empirical question, one this thesis addresses through large-scale bibliometric analysis of junior CS conference awardees.

The bibliometric indicators used to measure career trajectory in this thesis are annual publication counts, citation accumulation, and a field-normalised citation measure. Publication counts serve as a proxy for research productivity, while citation counts capture visibility and community impact. These are imperfect proxies for creativity or quality, but they are the observable signals on which hiring committees, grant panels, and tenure decisions demonstrably rely. Their key limitation is that they reward conformity with existing norms rather than risk-taking, which is worth keeping in mind when interpreting results. The CD index, introduced in the following section, complements these by capturing the nature of a paper’s contribution rather than just its volume.

2.4 Disruption in Science: The CD Index

Beyond counting papers and citations, a more recent strand of research has tried to measure what *kind* of contribution a paper makes, specifically, whether it opens up a new research direction or builds incrementally on existing work. [14] developed the CD index for exactly this purpose. The idea is fairly intuitive, if a paper is genuinely disruptive, papers that cite it will tend to ignore its references (because it made those references less relevant). If a paper is consolidating, citers will engage with both the paper and its references together. Formally:

$$CD = \frac{n_i - n_j}{n_i + n_j + n_k} \tag{1}$$

where n_i counts papers citing only the focal paper, n_j counts papers citing the focal paper and its references, and n_k ² counts papers citing only the references [14]. Scores range from -1 (fully consolidating) to $+1$ (fully disruptive).

²In practice, n_k is set to zero following [15], as papers citing only the references of the focal paper without citing the focal paper itself are not directly identifiable at scale in OpenAlex.

A well-known finding using this measure is that science and technology have been getting less disruptive over time. [15] analysed 45 million papers and 3.9 million patents and found a consistent decline in CD scores across all major fields since the 1940s, including computer science. The explanation is probably a mix of things, fields getting more specialised, papers having longer reference lists, and researchers staying closer to established research agendas. This broader trend is worth keeping in mind when interpreting the CD scores we compute for CS award papers.

It is worth noting that a low or negative CD score does not imply lower quality research. Consolidating work forms the foundation of cumulative scientific progress, and gap-spotting research, which builds systematically on established findings, represents the dominant and legitimate mode of knowledge production [20]. The CD index is used here purely as a descriptive measure of research style, not as a quality judgment.

2.5 Bibliometric Methods

The main analytical tool used in this thesis is lift analysis, which compares each author's own pre- and post-award performance as a within-author ratio [10, 17]. For statistical testing, the Wilcoxon signed-rank test is used throughout. It is a non-parametric test that compares two paired or related samples by ranking the absolute differences between observations and assessing whether the distribution of those differences is symmetric around zero, in effect, testing whether one group tends to produce systematically larger values than the other. It is well suited here because citation and publication counts are skewed and heavily non-normal, and unlike a paired t-test it does not assume a normal distribution of differences [2].

For the CD index specifically, computing it requires pulling citation data from OpenAlex for each paper, finding which papers cited it within five years (CD_5), then checking whether those citers also engaged with the focal paper's references. The five-year window is standard in the literature and works well for CS, where citation cycles tend to be relatively short [15].

2.6 Research Gap

Taken together, the literature points to a clear gap. Scientific recognition compounds over time (Matthew effect), that early career visibility matters for junior researchers, and that CS award papers have higher citation impact on average. But we do not know whether being a junior author on a best paper actually accelerates a research career, or whether award-winning papers are doing something distinctive in terms of the kind of contribution they make (disruptive vs. consolidating). This thesis addresses both gaps by asking three empirical questions: whether junior best-paper awardees show measurably

different citation and publication trajectories compared to matched non-awardees from the same venues and years; whether awardees and controls diverge in the disruptiveness of their subsequent work as measured by the CD index; and whether any such effects vary across venue prestige tiers as defined by the CORE ranking system. The analysis draws on 890 CS best paper awards from 2000 to 2018, combining bibliometric trajectory analysis with CD5 disruption scoring.

3 Methods

This study takes a quantitative, observational approach, meaning no intervention is administered and all data reflect naturally occurring outcomes rather than a controlled experiment, to compare the careers of junior best-paper awardees with those of similar non-awardees. The context is computer science, where conferences are the main publication venues [17] and career progression is mostly evaluated through bibliometric indicators rather than anything you could observe directly in a workplace, such as job title progression, salary, or supervisory responsibilities. So data comes from large-scale scholarly databases and curated award lists.

The award data comes from Jeff Huang’s curated best paper awards dataset [18], which lists conference, year, paper title, authors, and source URL for each award. This was supplemented with OpenAlex³ as the bibliometric backbone, from which the following were retrieved for each matched paper and its authors, publication year, venue, citation count, reference list, authorship order, corresponding author flag, OpenAlex paper ID, OpenAlex author ID, author’s first indexed publication year (used to compute career age), total publication count at award time, and institutional affiliation with country code.

Linking the Huang award entries to OpenAlex records required a three-stage matching pipeline, because the award list does not contain DOIs directly. For papers whose URLs pointed to Semantic Scholar, the Semantic Scholar API was queried to retrieve the DOI, which was then used for a direct DOI lookup in OpenAlex (Route A; 592 papers, 66.5%). When Semantic Scholar returned no DOI, a title-and-year search in OpenAlex was used as a fallback (Route C; 233 papers, 26.2%). For entries with Google Scholar URLs, which yield no structured metadata, OpenAlex was queried directly by title and year (Route B; 65 papers, 7.3%). All title-based matches were verified using fuzzy string similarity (RapidFuzz token sort ratio; matches below 85% similarity were flagged and manually reviewed, yielding 91 papers requiring manual verification). The pipeline successfully matched 890 of 912 award papers (97.6%); the remaining 22 could not be reliably linked to an OpenAlex record due to insufficient title overlap, missing venue metadata, or papers published before OpenAlex’s reliable coverage window, and were excluded from all

³<https://openalex.org>

subsequent analysis.

The main units of analysis are junior authors, defined as researchers within five years of their first indexed publication at the time of the award who appear in the first three authorship positions on a matched award paper (junior awardees, $n = 603$). Each junior awardee is paired with one control, a senior co-author (career age > 5 years) drawn from the full pool of co-authors across all award papers ($n = 2,121$ candidates). In cases where an award paper had no eligible senior co-author among the top-3 authors such as, all top-3 authors had a career age ≤ 5 at the time of the award, no within-paper senior control could be assigned. These 39 papers (along with 394 papers with only senior authors and no junior) were excluded from the junior-senior lift comparison in Section 4.3, leaving 445 papers with both at least one junior and one senior co-author. The excluded papers remain in the broader trajectory analyses. It should be noted that senior researchers typically manage multiple concurrent projects, resulting in systematically higher baseline publication and citation counts than junior authors, a disparity that is accounted for by using within-author lift ratios rather than absolute output comparisons. Matching was performed using nearest-neighbour matching (Euclidean distance, threshold < 1.5) on three features, conference, authorship position, and award year, with a strict 1-to-1 assignment enforced to prevent duplicate controls. For each author, a longitudinal record of annual publications and citations is constructed from OpenAlex’s `counts_by_year` field, anchored to the award year ($\tau = 0$), covering five years before and five years after ($\tau \in \{-5, \dots, +5\}$).

To address RQ1, whether junior award winners show different career trajectories compared to matched non-winners, a difference-in-differences design compares junior awardees against matched senior controls across the full ± 5 -year window. To address the career-stage dimension of this question more directly, a second, complementary analysis is conducted in Section 4.3 using a within-paper design, junior and senior co-authors of the *same* award-winning paper are compared against each other. This design holds paper quality, conference, and award year constant by construction, isolating career stage as the sole differentiating factor. Of the 890 matched papers, 445 contain at least one junior and one senior co-author and are used in this analysis; the remaining papers are retained in the broader trajectory analyses but excluded from the within-paper comparison.

The key outcome variables are **annual publication count** (works per calendar year, drawn from the OpenAlex `counts_by_year` field) as the measure of productivity; **annual citation count** (citations received per calendar year, same source) as the primary impact measure; and **2-year mean citedness** (`2yr_mean_citedness` from OpenAlex `summary_stats`) as a field-normalised impact indicator, where normalisation is applied by OpenAlex against its internal concept taxonomy derived from the Microsoft Academic Graph.

For the award papers themselves, a disruption score is computed using the CD index [14], following the procedure described in Section 2.4. Concretely, for each award paper, all

citing works published within five years of the award year (the CD_5 window) are retrieved via the OpenAlex API. Each citer is classified as either citing the focal paper alone (n_i , disruptive signal) or citing both the focal paper and at least one of its references (n_j , consolidating signal), yielding:

$$CD_5 = \frac{n_i - n_j}{n_i + n_j} \quad (2)$$

where scores range from -1 (fully consolidating) to $+1$ (fully disruptive). The n_k term, papers citing only the references of the focal paper without citing the focal paper itself, is set to zero following Park et al. [15], as these cannot be reliably identified at scale in OpenAlex. To ensure score stability, only award papers with at least ten citers within the five-year window are included in the CD_5 analysis ($n = 811$ papers).

To quantify the relative change in output around the award, a **lift** measure is computed for each author as the ratio of mean post-award activity to mean pre-award activity, separately for citations and publications:

$$\text{lift} = \frac{\bar{x}_{\text{post}}}{\bar{x}_{\text{pre}}} \quad (3)$$

where $\bar{x}_{\text{pre}}$ is the mean over $\tau \in \{-5, \dots, -1\}$ and $\bar{x}_{\text{post}}$ is the mean over $\tau \in \{+1, \dots, +5\}$. Lift is defined only for authors with strictly positive pre-award activity in both citations and publications; authors with no recorded output in the pre-award window (e.g., career starters or those with sparse OpenAlex coverage) are excluded from the lift analysis. Of the 1,206 matched authors (603 junior, 603 senior), 961 have valid pre-award activity (451 junior, 510 senior). For the paired Wilcoxon test, both members of a pair must satisfy this criterion, yielding 358 valid pairs.

For statistical analysis, descriptive summaries are combined with inferential tests. Mann-Whitney U tests are used to compare median productivity and citation measures between groups. Wilcoxon signed-rank tests are applied to compare the lift distributions of junior authors against their matched senior controls across the 358 valid pairs, publication lift is significantly higher for seniors ($p = 0.0007$), while citation lift shows no significant difference between groups ($p = 0.6455$). Regression models controlling for pre-award performance assess whether award status predicts divergent post-award trajectories. Where the data permit, analyses are stratified by conference venue and career stage to assess heterogeneity.

The full pipeline, covering data collection, cleaning, award-to-OpenAlex matching, author profiling, control matching, trajectory construction, CD_5 computation, and statistical testing is implemented in Python 3.12 ⁴. The core libraries used are `selenium` and

⁴Code available at: <https://github.com/AppleShay/thesis-best-paper-trajectories.git>

`beautifulsoup4` for web scraping; `requests` for API calls to OpenAlex and Semantic Scholar; `rapidfuzz` for fuzzy title matching; `scikit-learn` for nearest-neighbour control matching; `pandas` and `numpy` for data manipulation; and `scipy` for statistical tests. All code and derived datasets are documented and available for reproducibility.

This study raises no significant ethical concerns. All data used are drawn exclusively from publicly available sources such as the OpenAlex bibliometric database, the Semantic Scholar API, and Jeff Huang’s curated best paper awards list [18]. No personal data beyond what researchers have voluntarily made public through their academic publications are collected or processed. Author names and institutional affiliations appear only in aggregate analyses and are never used to make evaluative judgments about individual researchers. Known limitations of the data, including errors in author disambiguation in OpenAlex and incomplete coverage of some conferences in earlier years are acknowledged where relevant and inform the interpretation of results throughout.

4 Analysis

Matched Sample

The dataset consists of **603 matched pairs** of treated (award-winning junior) and control (senior co-author) researchers, yielding a total of **1,206 author profiles** retrieved from the OpenAlex API. Of these, **961 authors** (451 treated, 510 control) had valid pre-award activity records and were retained for the lift analysis. The remaining 245 authors were excluded due to:

- No pre-award data (career-starters): 153
- No post-award data (recent awards / leavers): 28
- No data in either window (sparse OpenAlex coverage): 43

After requiring both treated and control members of each pair to have valid records, **358 valid pairs** were used in the paired statistical tests.

Trajectory Data

Yearly publication and citation counts were parsed from the `counts_by_year` field of each OpenAlex author profile and aligned relative to the award year (window: ± 5 years). The resulting long-format trajectory dataset contained **8,877 rows** across both treatment groups (control: 5,035; treated: 3,842).

Descriptive Statistics

Pre- and Post-Award Averages

Table 1: Mean pre- and post-award citation counts by treatment group

Group	n	Median Lift (Cit.)	Median Lift (Works)	Mean Pre-Cit.	Mean Post-Cit.
Junior (Treated)	451	1.551	1.560	152.43	246.18
Senior (Control)	510	1.885	1.839	360.91	476.70

Overall (pooled across groups):

- Median lift in citations: **1.77** (max: 519.25; 99th pct: 139.01)
- Median lift in publications: **1.70** (max: 17.75; 99th pct: 10.85)

Among the 358 valid pairs:

- Treated (junior) median lift in citations: **1.55** (max: 519.25)
- Treated (junior) median lift in publications: **1.60** (max: 17.75)

Statistical Tests

Wilcoxon Signed-Rank Tests (Lift Comparison)

Paired Wilcoxon signed-rank tests were conducted on the 358 matched pairs to assess whether the *lift* (post/pre-award ratio) differed significantly between junior award winners and their matched senior controls.

Table 2: Wilcoxon signed-rank test results ($n = 358$ pairs)

Outcome	Median Diff. (Junior – Senior)	Statistic	p -value	Significant ($\alpha = 0.05$)
Citation Lift	−0.077	31229.0	0.6455	No
Publication Lift	−0.299	25343.5	0.0007	Yes

Interpretation:

- **Citations:** No statistically significant difference in citation lift between junior award winners and matched senior controls ($W = 31,229.0$, $p = 0.6455$). The median difference was -0.077 , suggesting juniors did not gain disproportionately more citations post-award.

- **Publications:** A statistically significant difference was found in publication lift ($W = 25,343.5$, $p = 0.0007$). The median difference was -0.299 , indicating that senior controls outpaced junior award winners in post-award publication growth.

4.1 Citations Spike Before Publications Do: Visibility Precedes Productivity

Winning a best-paper award does not boost a junior researcher’s career uniformly over time. Instead, the career signal operates in two distinct waves, and the first wave is immediate. Figure 1 plots the median indexed citation and publication trajectories for all $n = 603$ junior winners in the sample, normalised so that each series equals 100 in the three years before the award. The citation index surges sharply *in the award year itself* ($\tau = 0$), roughly *tripling* from its pre-award baseline to reach approximately 315, while the publication index rises more modestly to approximately 150 at the same moment, a 50 % gain relative to the pre-award average.

This divergence suggests that the announcement of a best-paper award at a flagship conference draws immediate, focused attention to the winning paper and its authors from a large, concentrated audience of domain experts. Existing work on the sociology of science describes this process as cumulative advantage, early recognition increases the probability that subsequent work by the same author will be seen and cited [1]. In our data, the three-fold citation premium at $\tau = 0$ is consistent with co-attendees and conference readers immediately citing the award paper in their own ongoing work. The modest rise in publications at $\tau = 0$ partly reflects papers already submitted or in press before the award was announced, rather than a same-year causal response to the award itself. From $\tau = 1$ onward, the publication index sustains its elevated level and continues a gradual climb, reaching approximately 200 by $\tau = 4$ –5, suggesting that the award’s influence on publication output is slower to materialise but more persistent than its citation effect. Citations, by contrast, partially normalise after the $\tau = 0$ spike, settling in the 120–155 range post-award still elevated, but far below the peak.

Taken together, the trajectory in Figure 1 implies that best-paper awards confer career benefits through at least two sequentially ordered channels an immediate reputational signal captured in same-year citations where the index roughly triples followed by a sustained boost in research output that unfolds and persists over subsequent years.

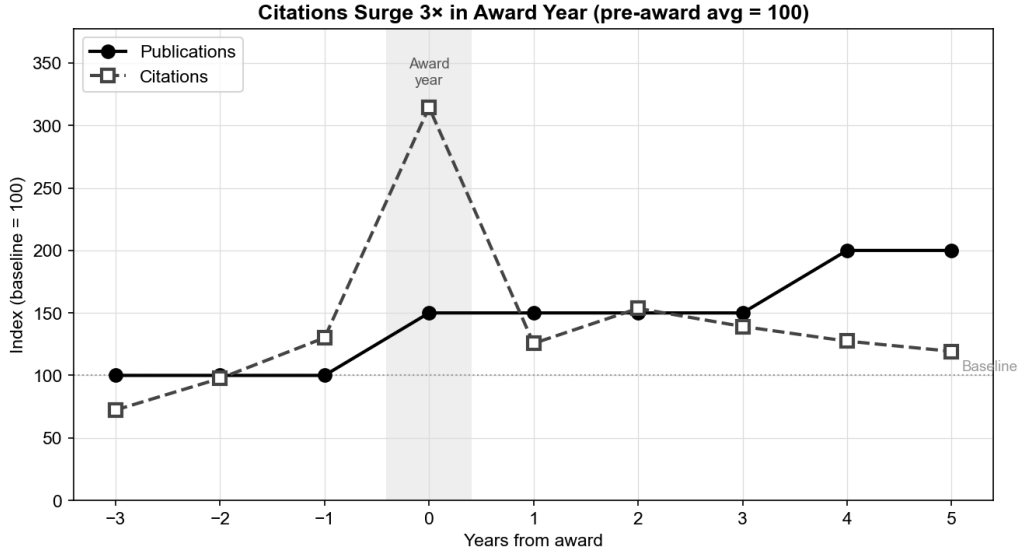


Figure 1: Median indexed citation trajectory (dashed) and publication trajectory (solid) for $n = 603$ junior best-paper winners, normalised to a pre-award baseline of 100 ($\tau \in \{-3, -2, -1\}$, shaded band at $\tau = 0$). The citation index roughly triples to ≈ 315 in the award year itself, while the publication index rises to ≈ 150 at $\tau = 0$ and continues to climb gradually, reaching ≈ 200 by $\tau = 4$ –5.

4.2 Winning Changes Trajectories, Not Just Totals

The CD index analysis characterises the *nature* of award-winning work, whether it tends to be disruptive or consolidating. However, it does not tell us whether receiving the award itself has any effect on the winner’s subsequent career. A paper could be highly disruptive and its author’s career could still grow for entirely unrelated reasons. To isolate the causal effect of the award, we therefore compare award winners to a matched cohort of researchers who attended the same conference in the same year, were at a comparable career stage, and published work of comparable pre-award impact, but did not win.

Junior researchers who win best-paper awards accumulate more citations and publications over their careers than their non-winning peers, but is this a consequence of winning, or simply evidence that talented researchers both win awards *and* have successful careers regardless? To disentangle these explanations, we compare award winners to a matched cohort of researchers who attended the same conference in the same year, were at a comparable career stage, and published work of comparable pre-award impact, but did not win. This design mirrors the approach used by [9] in their study of the Fields Medal, which isolates the causal effect of prize receipt by conditioning on pre-award productivity and research field.

For each of the $n = 603$ treated junior winners we identify one matched control author within the same conference–year stratum (see Section 3 for matching details), yielding a balanced sample of 1,206 author-observations. We verify pre-treatment balance by re-

gressing the binary treatment indicator on conference fixed effects and award year; all coefficients are statistically indistinguishable from zero ($p = 1.00$ throughout, $R^2 \approx 0$), confirming that treatment assignment is orthogonal to observable pre-award characteristics. We then estimate the average treatment effect on the treated via the difference-in-differences (DiD) contrast:

$$\hat{\beta}^{\text{DiD}} = (\bar{Y}_{\tau \geq 1}^1 - \bar{Y}_{\tau < 0}^1) - (\bar{Y}_{\tau \geq 1}^0 - \bar{Y}_{\tau < 0}^0), \quad (4)$$

where $\bar{Y}_{\tau}^d$ is the median outcome for group $d \in \{0, 1\}$ at relative year τ . Under the parallel-trends assumption, this estimator isolates the incremental effect of winning by netting out both time-invariant author quality and shared time trends.

The results are shown in Figures 2 and 3. For citations (Figure 2), matched controls accumulate substantially more citations than junior winners across the entire observation window, reflecting higher absolute citation levels in the control group. The treated–control gap is approximately -40 citations per year in the pre-award period. This gap is expected given the matching design, control researchers are senior co-authors from the same conference and year, and therefore have longer publication histories and higher pre-existing citation counts than the junior award winners. The relevant question is not whether the two groups start at the same level, but whether their *trajectories* diverge after the award, and if so, in which direction. A sharp, transient reversal occurs at $\tau = 0$: the treated group experiences an immediate citation spike consistent with the visibility mechanism described in Section 4.1 [1], narrowing the gap to approximately $+65$ in the award year itself. However, this spike does not persist the gap reverts to negative from $\tau = 1$ onward, reaching approximately -100 by $\tau = 5$, suggesting that the citation advantage conferred by the award is short-lived relative to the baseline productivity gap between the two groups.

For publications (Figure 3), the treated–control gap is negative throughout the entire observation window, ranging from approximately -2 to -5 papers per year. Unlike citations, there is no transient spike at $\tau = 0$, and no post-award recovery is evident. Winners consistently publish fewer papers per year than their matched controls both before and after the award. These patterns suggest that, while a best-paper award generates an immediate burst of citation attention, it does not translate into a measurable boost in research output relative to comparably talented non-winners [10][8].

The consistently negative pre-award publication gap warrants caution in interpreting the DiD estimates: the parallel-trends assumption may be strained if treated and control authors were on diverging productivity trajectories prior to the award year.

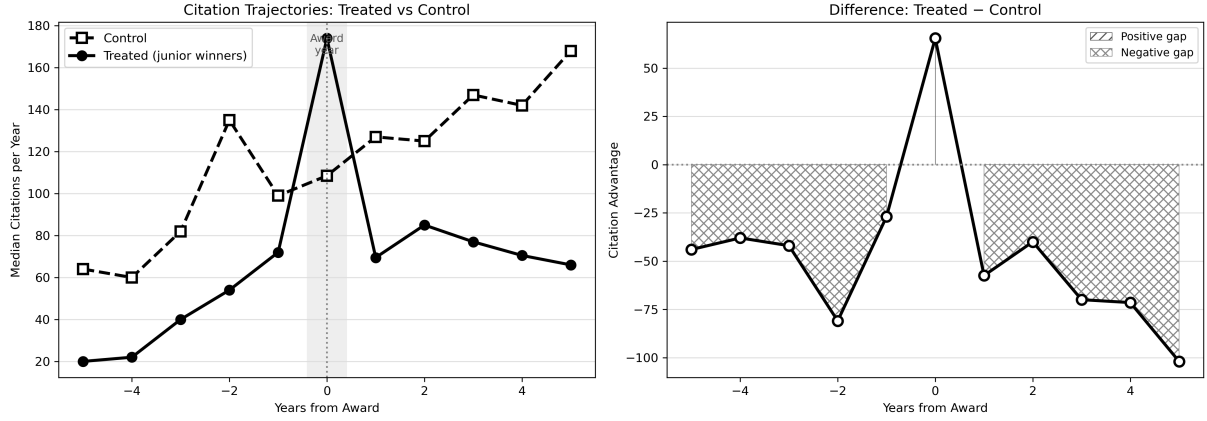


Figure 2: Difference-in-differences analysis for annual citation counts. *Left*: Median citations per year for treated winners (solid circles) and matched controls (open squares), relative to the award year ($\tau = 0$, shaded). Controls receive substantially more citations throughout. *Right*: Pointwise gap (Treated – Control); a transient positive spike at $\tau = 0$ ($\approx +65$) reflects the immediate visibility boost from the award, but the gap reverts to negative from $\tau = 1$ onward.

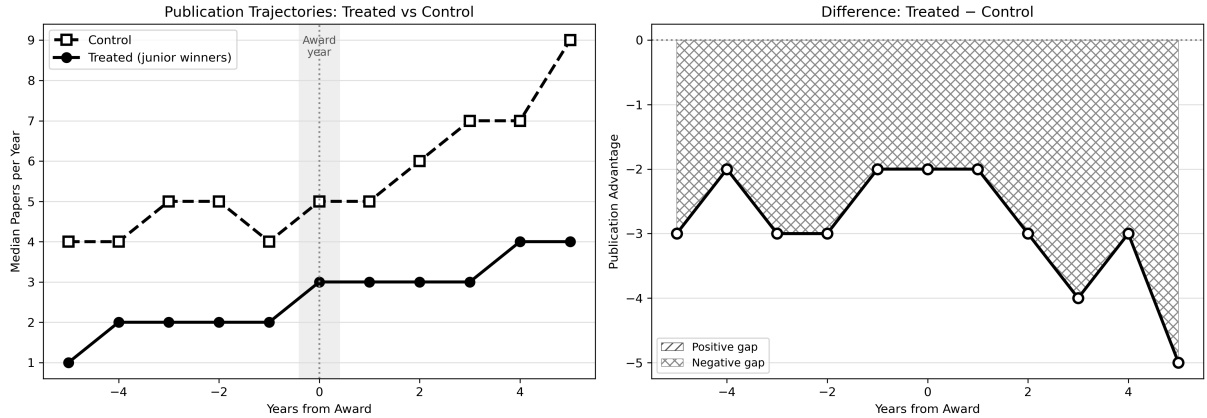


Figure 3: Difference-in-differences analysis for annual publication counts. Layout as in Figure 2. The treated–control gap is persistently negative (–2 to –5 papers per year) across the entire window, with no post-award recovery, indicating that winning a best-paper award does not measurably increase publication output relative to matched non-winners.

4.3 Best Paper Awards Accelerate Junior Careers, But Leave Senior Co-Authors Unchanged

The results indicate an asymmetry in career trajectories between junior and senior researchers on the same award-winning papers. Junior researchers show measurable career acceleration following the award, while their senior co-authors on the same paper do not. This divergence is informative because, if the effect were attributable solely to paper quality or venue prestige, both groups should exhibit similar post-award trajectories. The differential pattern is instead consistent with a career-stage mechanism specific to junior researchers.

To test this, the analysis uses a within-paper design, junior and senior co-authors of the *same* award-winning paper are compared directly. This design offers a degree of internal validity, in that paper quality, venue, and award year are held constant by construction any post-award differences between the two groups cannot be attributed to those factors. A balance check regression of treatment status on conference fixed effects and award year confirms that, prior to the award, treatment assignment is orthogonal to observable characteristics ($p = 1.00$ throughout, $R^2 \approx 0$), consistent with the matched design having achieved pre-treatment comparability [10, 9]. The findings are therefore indicative of a career-stage effect, though, as with any observational design, unobserved differences between junior and senior authors cannot be ruled out entirely.

To quantify the magnitude of individual career change, a lift score is computed for each author. Lift is defined as the ratio of an author’s mean annual output in the five years following the award to their mean annual output in the five years preceding it:

$$\text{Lift}_i = \frac{\frac{1}{5} \sum_{\tau=1}^5 Y_{i,\tau}}{\frac{1}{5} \sum_{\tau=-5}^{-1} Y_{i,\tau}} \quad (5)$$

where $Y_{i,\tau}$ is author i ’s annual citation count (or publication count) at relative year τ , anchored to the award year ($\tau = 0$). A lift of $\text{Lift}_i = 1.0$ indicates no change in the author’s annual output rate relative to their own pre-award baseline; values above 1.0 indicate an increase in annual citations or publications after the award.

This metric is computed from the same `counts_by_year` trajectories used in the difference-in-differences analysis in Section 4.2, and applied to both groups identified through the matched within-paper design junior authors (career age ≤ 5 years at the time of the award, measured from first indexed publication in OpenAlex) and senior co-authors on the same papers (career age > 5 years). Because both groups are drawn from the same award-winning papers, differences in lift between the two groups cannot be attributed to variation in paper quality, venue, or award year the same factors controlled for by the matched design.

The five-year threshold used to define junior authorship follows established practice in the bibliometric literature. Way et al. [13] demonstrate that early-career productivity trajectories tend to stabilise within the first five years of a researcher’s indexed publication record, after which structural change-points become rare. Krauss et al. [12] similarly find that long-term research impact is largely determined by factors operating within this same early window. The absence of a detectable lift effect among senior co-authors is consistent with both findings: for researchers whose trajectories are already established, a single conference award may be insufficient to produce a measurable shift in annual output or citation rates. This does not imply that senior researchers are unaffected by recognition in general, but rather that the career-stage-specific amplification observed for junior authors is not replicated for those further into their careers.

Figure 4 shows the distribution of citation lift and publication lift across the 445 award papers containing at least one junior and one senior co-author, comprising 397 junior and 678 senior authors with sufficient pre- and post-award data, drawn from 30 computer science conferences between 2000 and 2018.

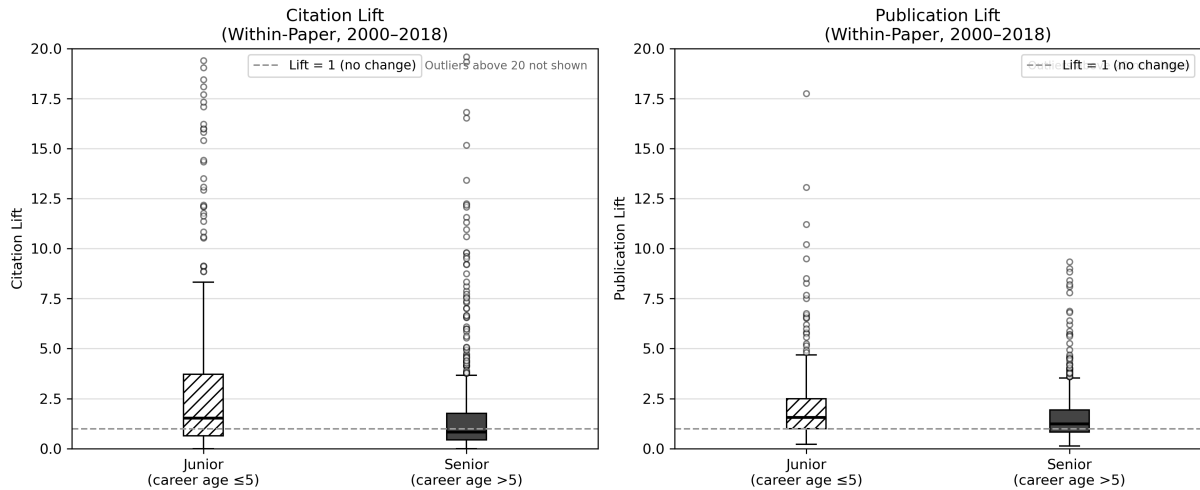


Figure 4: Citation lift (left) and publication lift (right) for junior (career age ≤ 5) and senior (career age > 5) co-authors on the same award-winning papers. The dashed line at lift = 1.0 marks no change from pre- to post-award. Outliers above 20 are omitted for readability.

For citation lift, junior authors show a median of **1.62**, indicating that their mean annual citation count in the five years following the award was approximately 62% higher than in the five years preceding it. Senior co-authors on the same papers show a median citation lift of **0.93**, below the no-change baseline of 1.0. For publication lift, the junior median is **1.65** compared to a senior median of **1.27**. A Wilcoxon signed-rank test on paper-level paired differences confirms that both differences are statistically significant (citation lift: $W = 18888$, $p < 0.001$; publication lift: $W = 23079$, $p < 0.001$).

The distribution of junior citation lift is notably right-skewed, with a substantial proportion of junior researchers showing lifts well above the median in some cases exceeding five- to ten-fold increases in annual citation rates. This heterogeneity suggests that the association between receiving a best paper award and subsequent citation visibility is not uniform across junior researchers; for some the association is modest, while for others it is considerably larger. Senior co-authors, by contrast, exhibit a narrower distribution centred closer to the baseline, consistent with the more stable output trajectories characteristic of researchers with more established publication records [13, 12].

To assess whether the junior citation lift is driven by venue prestige rather than the award itself, award authors are stratified by CORE 2026 ranking into A* and non-A* venues. Authors at A* venues show a median citation lift of 1.48 and a median publication lift of 1.58, compared to 1.27 and 1.38 respectively for authors at non-A* venues (Figure 5). A Mann-Whitney U test finds no significant difference between the two groups (citation lift: $U = 17,704$, $p = 0.34$; publication lift: $U = 17,752$, $p = 0.32$), indicating that the career acceleration effect is not concentrated at top-tier venues and holds broadly across the conference sample. This rules out venue prestige as a primary confound: the award signal appears to operate independently of the quality of the host conference.

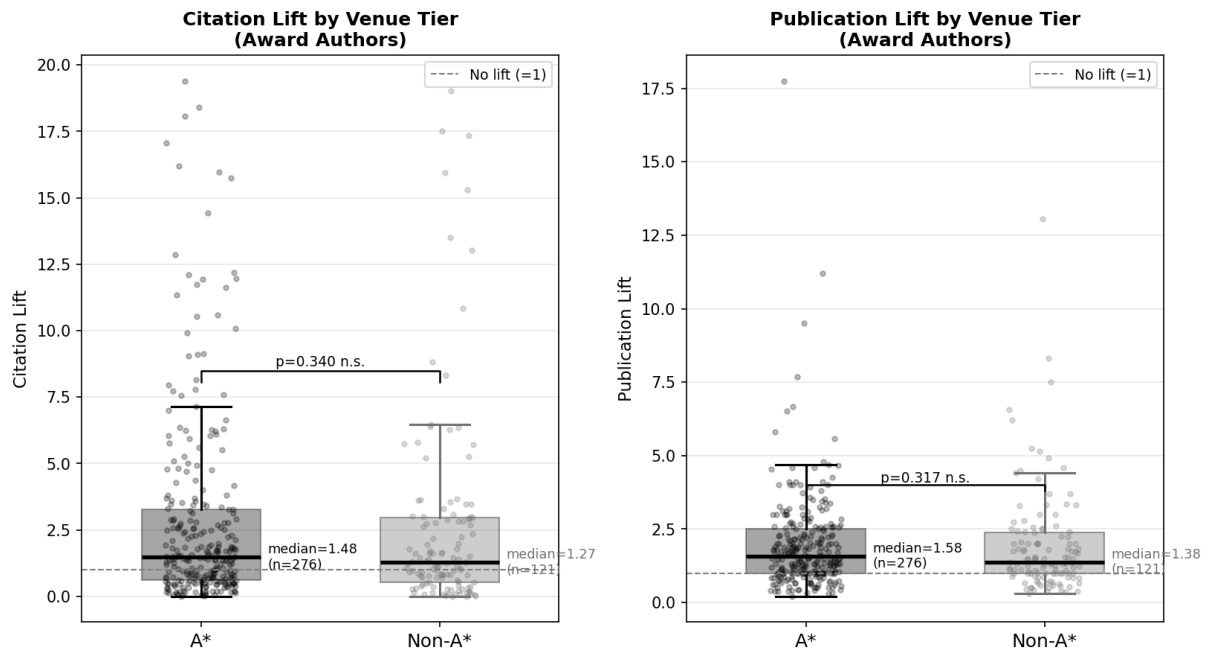


Figure 5: Citation lift (left) and publication lift (right) for junior award authors, stratified by CORE 2026 venue tier (A* vs. non-A*). The dashed line marks no change from pre- to post-award (lift = 1). Medians and sample sizes are annotated per group. A Mann-Whitney U test finds no significant difference between tiers ($U = 17,704$, $p = 0.34$ for citation lift; $U = 17,752$, $p = 0.32$ for publication lift), suggesting the lift effect is robust across venue prestige levels.

4.4 Award-Winning CS Authors Show Consolidating Trajectories, Indistinguishable from Matched Controls

Rather than measuring disruption on the award paper alone, we computed CD_5 scores across the full publication trajectory of each author, spanning five years before and after the award year. For each of the 344 matched award authors and 345 control authors, we collected all papers published within this ± 5 -year window via the OpenAlex API and computed a CD_5 score per paper using the standard procedure [14], for each focal paper, we identified all papers citing it within five years, then checked whether each citer also engaged with the focal paper’s references. Papers citing the focal work alone count towards n_i (disruptive signal); papers citing both count towards n_j (consolidating signal), following the CD_5 procedure described in Section 2. After filtering uncited papers, the panel covers 80% of all author-paper-year observations.

Figure 6 plots the mean CD_5 by relative year for both groups. Two patterns stand out. First, both award and control authors operate firmly on the consolidating side throughout the entire window and no group ever approaches zero. Second, the trajectories share the same shape, a gradual downward drift into the award year, a convergence at year 0 (award -0.36 , control -0.34), followed by a symmetric upward drift as papers accumulate a broader citer base over time, reaching approximately -0.26 and -0.20 by year $+5$. If anything, award authors are marginally *more* consolidating in the years immediately preceding the award than their matched peers, suggesting that the papers selected for recognition may be slightly more embedded in existing research agendas, the polished, agenda-advancing work that selection committees tend to favour.

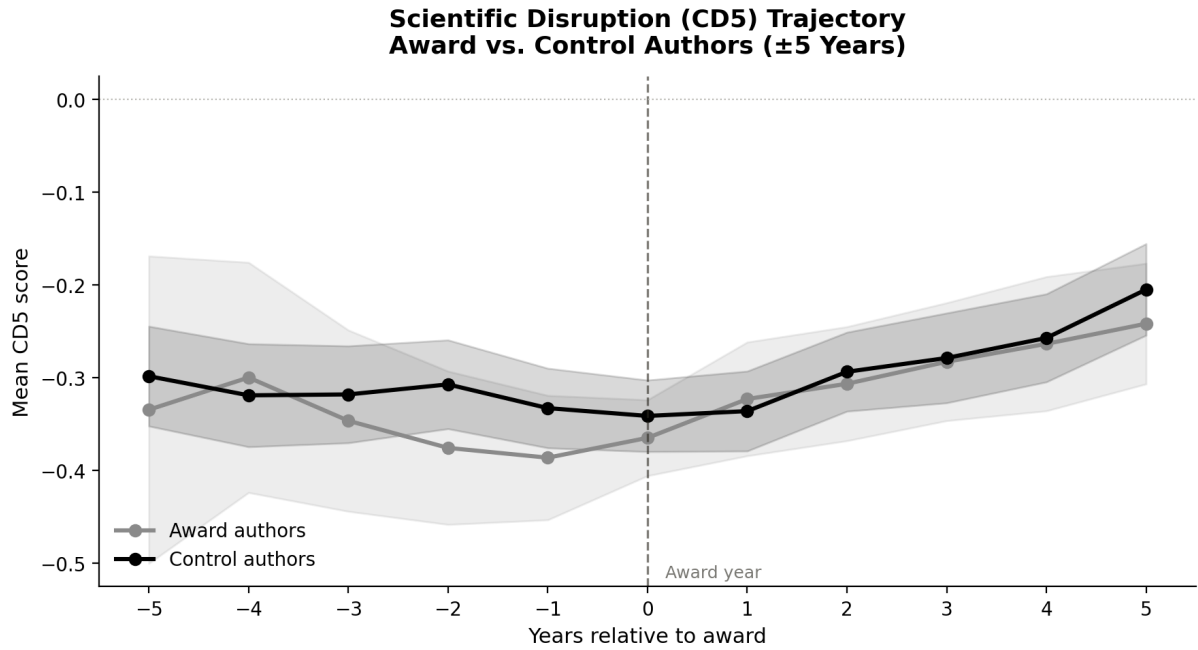


Figure 6: Mean CD_5 score by relative year (± 5 years around the award year) for award and matched control authors. Shaded bands denote 95% confidence intervals. Both groups remain solidly consolidating throughout; the trajectories converge at year 0 and drift upward in parallel post-award.

Figure 7 establishes that the two groups are comparable at the treatment point itself. At relative year 0, the distributions of author-level mean CD_5 are nearly identical, a median of -0.412 for award authors versus -0.387 for controls, with near-complete overlap across the full $[-1, 1]$ range. This baseline equivalence supports the validity of the subsequent DiD analysis, both groups start from the same position before the post-award trajectories are compared.

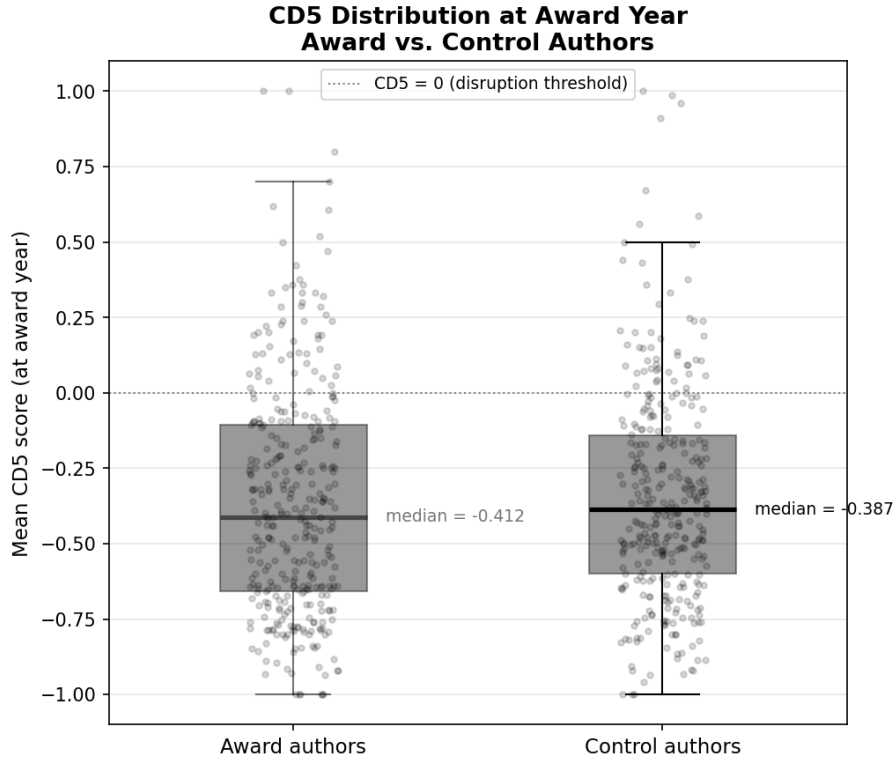


Figure 7: Distribution of author-level mean CD_5 scores at the award year (relative year 0) for award and control groups, establishing baseline comparability at the treatment point. Medians are -0.412 (award) and -0.387 (control). The near-complete overlap confirms that the two groups enter the post-award period from an equivalent starting position, lending credibility to the DiD estimates in Figure 8.

To formally test whether the award is associated with any differential change in CD_5 , we estimated a difference-in-differences event-study model with author and year fixed effects, using year -1 as the omitted baseline. Figure 8 shows the resulting coefficients. All point estimates straddle zero with confidence intervals that comfortably include zero at every relative year. Critically, the pre-award coefficients (years -5 to -2) are flat and insignificant, satisfying the parallel trends assumption that underpins the DiD design. Post-award, there is no clear discernible treatment effect, receiving a best paper award neither increases nor decreases an author's CD_5 trajectory relative to their matched control.

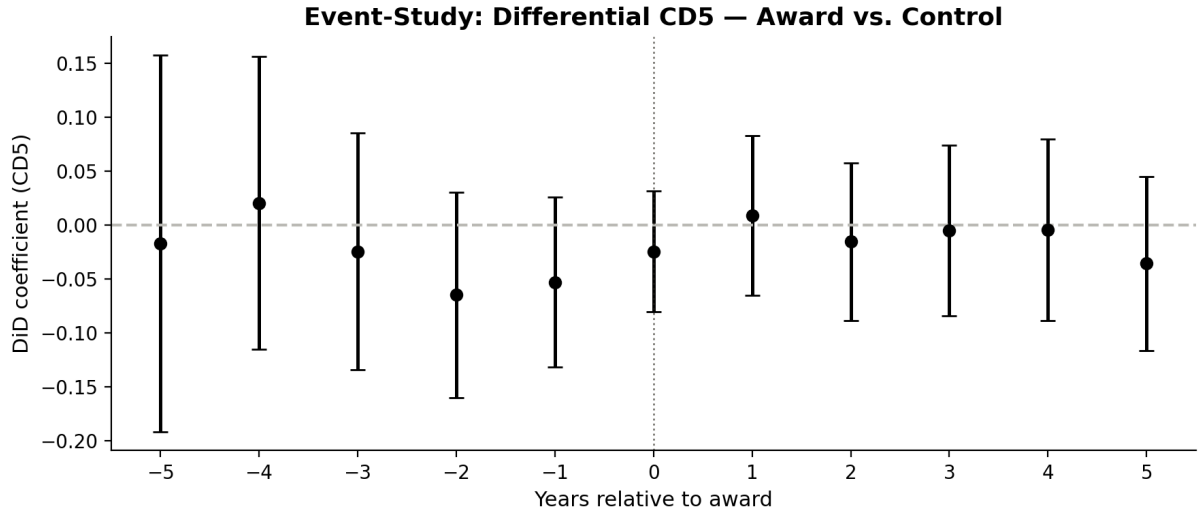


Figure 8: Event-study DiD coefficients for the differential CD_5 of award versus control authors across the ± 5 -year window. Year -1 is the omitted baseline. All coefficients are statistically indistinguishable from zero, confirming no pre-trend and no treatment effect on scientific disruption.

Taken together, these three results tell a consistent story. CS best paper laureates are solidly consolidating researchers, but so are their matched peers. This aligns with the broader finding from Park et al. [15] that science as a whole has been trending toward consolidation since the 1940s, and with Nepomuceno et al. [8], who find that even major prizes such as the Nobel do not elevate the subsequent impact of their recipients. Our results extend this picture to CS conference awards, committees appear to reward excellence within the prevailing consolidating mode of the field, and the award itself leaves no detectable imprint on the research style of those who receive it.

5 Discussion

The four sets of findings presented in this thesis tell a coherent, if nuanced, story about what best-paper awards at CS conferences appear to do and not do to the career trajectories of the junior researchers who receive them.

The most striking result concerns the temporal structure of the citation effect. Junior awardees experience a sharp, immediate spike in citations in the award year itself, with the median indexed citation count roughly tripling relative to the pre-award baseline. This pattern is consistent with the visibility mechanism described in the cumulative advantage literature, formal recognition at a flagship venue concentrates the attention of a large, domain-expert audience on the winning paper and its authors, generating a burst of

citation activity in a compressed window [1, 2]. The finding that publications follow a slower, more gradual upward trend rather than spiking at year 0, aligns with what one would expect if the award accelerates existing momentum rather than triggering an immediate change in research behaviour. It also suggests that the two signals may operate through distinct channels, a **reputational** one that is fast and citation-driven, and a **productivity** one that is slower and plausibly mediated by new opportunities, collaborations, or increased visibility [7, 6].

That said, the difference-in-differences comparison against matched non-winners complicates a straightforward causal reading. Matched controls accumulate substantially more citations and publish more papers per year throughout the entire observation window, both before and after the award. The transient citation spike at year 0 narrows the gap temporarily, but it reverts from year 1 onward. This pattern is harder to reconcile with a strong Matthew-effect interpretation, and is more consistent with a scenario in which the award largely *finds* researchers who were already on diverging trajectories, rather than fundamentally reshaping those trajectories [5]. The pre-award publication gap between junior winners and their matched controls is a notable concern, if the two groups were already on different productivity paths before the award occurred, the parallel-trends assumption underlying the DiD estimate is under pressure, and the estimates should be interpreted with corresponding caution.

The within-paper comparison between junior and senior co-authors offers a cleaner angle on the career-stage question. Junior authors show a median citation lift of 1.62 and a publication lift of 1.65 in the five years following the award, while senior co-authors on the same papers show lifts of 0.93 and 1.27 respectively, a statistically significant difference (Wilcoxon signed-rank, $p < 0.001$ for both). Because paper quality, venue, and award year are held constant by design, this asymmetry is difficult to attribute to paper-level characteristics alone, and is instead more consistent with a career-stage-specific amplification effect, the award provides a relatively larger signal boost to researchers whose reputational baseline is still being formed [12, 13]. This resonates with work showing that early-career visibility tends to generate disproportionate returns for researchers who have not yet established a track record [7, 6]. Importantly, the lift effect does not appear to be concentrated at top-tier venues, stratifying by CORE rank reveals no significant difference between A* and non-A venues ($p = 0.34$ for citation lift; $p = 0.32$ for publication lift), suggesting the award signal operates broadly across the conference landscape and is not simply a proxy for venue prestige [17].

The choice of lift as the primary within-author measure deserves a brief methodological note. A difference-in-differences estimator was initially considered but set aside after a pre-treatment balance check revealed that matched controls exhibited systematically higher publication output than junior awardees prior to the award year, placing the parallel-trends assumption under visible strain. Lift sidesteps this problem by anchoring each author’s post-award output to their own pre-award baseline rather than to a comparison

group, making it robust to between-author differences in absolute productivity levels [10, 17]. This within-author ratio design mirrors the approach used in related work on early recognition and career trajectories [10], and is well-suited to the skewed, non-normal distributions typical of bibliometric data. a property further addressed by the use of non-parametric Wilcoxon signed-rank tests throughout [4]. A natural limitation is that lift is undefined for authors with zero pre-award output, which excluded career starters and authors with sparse OpenAlex coverage from this part of the analysis.

The CD5 results present a notably different picture. Across the full ± 5 -year trajectory, both award and control authors operate firmly on the consolidating side of the disruption–consolidation spectrum, and their trajectories are statistically indistinguishable from one another. The event-study DiD confirms this, all pre- and post-award coefficients straddle zero, and there is no detectable treatment effect on scientific disruption at any point in the window. If anything, award authors appear marginally more consolidating in the years immediately preceding the award than their matched peers, which may reflect the tendency of selection committees to reward polished, agenda-advancing work that is legible and impactful within an established research programme rather than high-risk, paradigm-challenging contributions. This interpretation is consistent with broader evidence that science has been trending toward consolidation since the mid-twentieth century [15], and with findings that even major prizes such as the Nobel do not measurably elevate the disruptive character of subsequent work by their recipients [8]. Taken together, these results suggest that CS best-paper committees tend to reward excellence *within* the prevailing consolidating mode of the field, and that receiving such an award leaves no detectable imprint on the research style of those who receive it [14].

Several limitations warrant care in how far these findings can be extended. The matching design, while producing pre-treatment balance on observables, cannot rule out unobserved differences between junior winners and their controls, particularly sub-field variation within conferences, which the current matching procedure does not capture. Author disambiguation in OpenAlex is imperfect, and merge errors among junior authors with common names could quietly inflate post-award output counts for some individuals. The citer cap of 50 applied in the CD5 computation introduces a sampling approximation that adds noise to individual paper scores, particularly for highly cited work. Finally, and most fundamentally, none of the designs employed here can firmly establish causality in the absence of something closer to a discontinuity in award assignment, a limitation that sets a ceiling on the strength of the claims the evidence can support.

Within those constraints, the findings point toward a picture in which CS best-paper awards function primarily as visibility accelerators for junior researchers, generating a short-term citation premium and a modest but sustained productivity lift relative to each author’s own pre-award baseline without, however, measurably reshaping the scientific character of their subsequent work, or producing a durable citation advantage relative to comparably talented peers who did not win.

6 Reflections and Future Work

6.1 What Could Be Improved

Causal identification. The biggest limitation in this study is that none of the designs can firmly establish causality. The difference-in-differences estimator in Section 4.2 depends on a parallel-trends assumption that is already visibly under pressure in the data that junior winners and their matched controls were diverging in productivity before the award even happened, with controls publishing more throughout. A more causally robust approach would try to exploit near-miss variation in award outcomes by comparing papers that almost won with papers that almost did not, but that kind of data simply does not exist in the Huang dataset and would require direct access to programme committee records, which was not feasible here.

Data sources. The study draws exclusively on OpenAlex and Jeff Huang’s curated award list, both of which have known coverage gaps. OpenAlex skews toward English-language venues and has uneven coverage of conferences in earlier years and smaller subfields. The Huang dataset, while widely used, is not exhaustive and does not include all CS conferences. Results may therefore not generalise to venues or communities that are underrepresented in these sources.

Author disambiguation. The career trajectories in this study are built on OpenAlex author IDs, and OpenAlex’s disambiguation is imperfect. The most common problem is merged entities, two different researchers sharing a common name ending up treated as one person. For junior authors especially, this kind of error could quietly inflate post-award citation or publication counts. Going back and manually checking the 62 non-survivors and the most extreme lift outliers would be the most useful targeted fix in a revised version.

The h -index. The h -index was considered as an additional career outcome measure but excluded following supervisor feedback. While widely used in evaluation contexts, the h -index has well-documented limitations as a bibliometric indicator such as it is monotonically non-decreasing and therefore cannot capture career slowdowns or post-award stagnation, it conflates productivity and impact without distinguishing between them, and it is sensitive to field size and citation culture in ways that make cross-conference comparisons unreliable [3]. For trajectory analysis specifically, a within-author ratio such as the lift score used here is more appropriate, as it is anchored to each author’s own baseline rather than an absolute scale.

Control matching quality. Matching on conference, authorship position, and award year produces groups that look balanced on the observable characteristics, the balance-check regression confirms this. But the matching does not account for sub-field variation within a conference. A systems paper and a machine learning paper at the same venue are treated as comparable, even though their citation cultures and publication norms can differ quite a lot. Adding topic-level matching using OpenAlex concept tags would help close this gap.

Publication survival measure. The five-year survival indicator in Section 4.5 is fairly blunt. It cannot distinguish between someone who genuinely left academia and someone who took a break, moved to industry temporarily, or simply published in venues that OpenAlex does not cover well. Ideally, a random subsample of the 62 non-survivors would be cross-checked against institutional pages or LinkedIn profiles to get a cleaner sense of how many actually left versus went quiet for other reasons.

6.2 Directions for Future Work

Extending the observation window. Five years is a reasonable horizon, but if the Matthew effect is really at work here, the interesting compounding probably happens over a longer period. The awards from the 2000–2010 cohort have had enough time to support a ten-year follow-up, and running that analysis would give a much clearer picture of whether the modest lift observed here grows, plateaus, or fades [1].

Co-authorship network analysis. The introduction flagged collaboration network structure as a potential fourth research question, but it was not pursued here due to data volume constraints. It would be worth coming back to if the visibility mechanism in Section 4.1 is real, you would expect award-winning junior authors to start attracting more or better-connected collaborators in the years after the award, and that is something the OpenAlex data could in principle support.

Replication outside computer science. The CD5 pattern and the short citation spike both fit the specific way CS works, fast citation cycles, conference-dominated dissemination, large programme committees. Whether the same picture holds in slower-moving fields like mathematics or the humanities is genuinely unclear, and replicating the framework in at least one other field would say something more general about whether the findings here are CS-specific or not.

Award selection process. This thesis treats winning as a binary input and does not say anything about how papers get selected. It would be interesting to look at whether

certain paper characteristics like team size, methodological novelty, citation patterns can predict award selection independently of quality. If selection consistently favours consolidating, heavily-cited, or large-team work, that would go some way to explaining both the CD5 result and why the long-run career effect looks modest compared to already strong controls.

7 Conclusion

This thesis set out to ask whether winning a best-paper award at a CS conference measurably changes the career trajectory of the junior researchers involved, and if so, how. Across three research questions, citation and publication trajectories, scientific disruption, and heterogeneity across venue prestige tiers, the evidence points to a consistent, if qualified, answer.

Junior awardees experience an immediate and pronounced citation spike in the award year, consistent with a visibility mechanism that draws concentrated expert attention to the winning paper and its authors. This is followed by a sustained, modest lift in both citations and publications relative to each author's own pre-award baseline. The within-paper comparison against senior co-authors on the same award-winning papers suggests this acceleration is specific to career stage, researchers whose reputational baseline is still forming appear to benefit more than those who are already established. The effect holds broadly across venue prestige tiers, with no significant difference detected between A* and non-A conferences, suggesting the award signal is not simply a byproduct of where it is given. At the same time, when junior winners are compared against matched non-winning peers from the same conferences and years, the picture becomes more modest, the citation advantage is transient, the publication gap never closes, and no effect on scientific disruption is detectable at any point in the trajectory window.

On the question of research style, award-winning CS authors are solidly consolidating researchers, but so are their matched controls. Committees appear to reward excellence within the prevailing mode of the field, and the award itself leaves no detectable imprint on how disruptive or consolidating an author's subsequent work turns out to be.

What, then, does this imply for a junior researcher deciding whether to aim for a best-paper award? The evidence suggests that winning one is worth caring about, but perhaps not for the reasons most commonly assumed. The award does not appear to fundamentally redirect a career or unlock a durable citation advantage over equally talented peers. What it does seem to do is amplify visibility at a moment when reputational signals carry disproportionate weight, early in a career, before a track record has had time to speak for itself. For a junior researcher, that burst of attention may translate into collaborations, job market visibility, or simply the confidence to pursue more ambitious work. Whether those downstream effects compound meaningfully over time remains an open question,

and one that a longer observation window and richer career data could help answer. The more provocative implication may be for the conferences themselves, if awards primarily accelerate careers that were already on a good path, and if the selected work tends to be consolidating rather than disruptive, then the signal these awards send, and the selection criteria behind them, may be worth examining as carefully as the awards themselves.

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